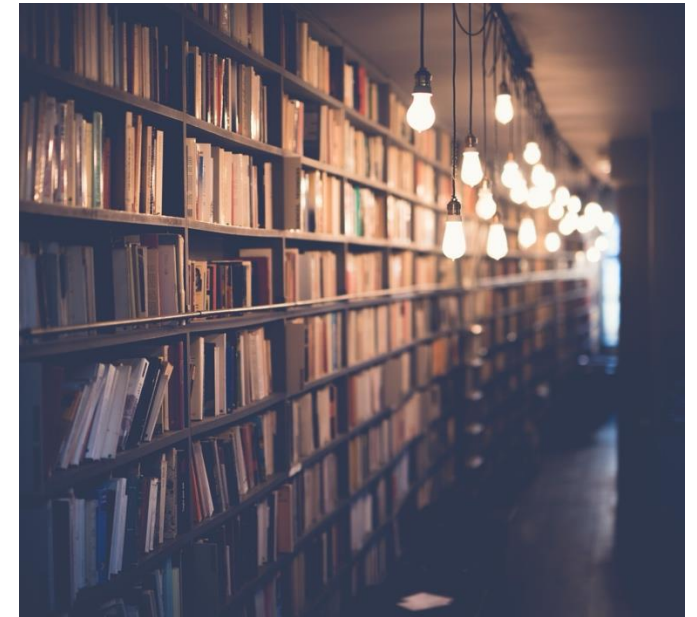




UiO : **CEMO** – Centre for Educational Measurement
University of Oslo

Meta-Analysis in Education

From Theory to Practice with R:
Multivariate Meta-Analysis and MASEM



Ronny Scherer & Diego G. Campos

Workshop at the TU Dortmund
21.02.2025



BEGIN.

Multivariate Meta-Analysis

Model correlated effect sizes explicitly and go beyond a constant rho

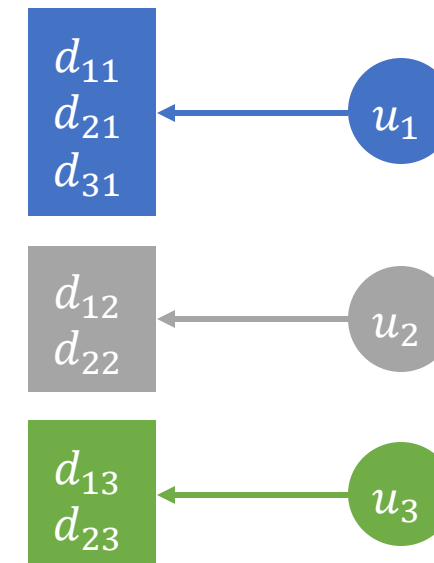
Multivariate Meta-Analysis (MVMA)

Model the correlated data structure

- Effect sizes y_{ij} , $i = 1, \dots, k; j = 1, \dots, m$
- Sampling variances **and** covariances v_{ij}
- Multiple correlated effect sizes nested in studies (usually from multiple measures of the same sample)
- **Correlated effects**

Main idea

Incorporate the sampling covariances in the model with multiple effect sizes



(Van den Noortgate et al., 2015)

Activity: Find Examples.

Examples

Describe an example of a meta-analytic data set, for which correlated effects could be assumed.



Multilevel vs. Multivariate Meta-Analysis

Key similarities and differences

MLMA

- Reported effect sizes are nested in studies (different samples)
- Sampling errors are conditionally independent
- 3LREM special case of MVREM with no correlation between effects
- **Hierarchical effects**

MVMA

- Multiple effect sizes from the same participants (e.g., multiple treatment or endpoints)
- Sampling errors are dependent
- MVREM approximated by 3LREM
- Sampling covariances must be known
- **Correlated effects**

Multivariate Meta-Analysis (MVMA)

Multivariate random-effects model (MVREM)

Part 1: $y_i = f_i + e_i$
 $e_i \sim MVN(0, V_i)$

Part 2: $f_i = \beta_R + u_i$
 $u_i \sim MVN(0, T^2)$

Total:

$$y_i = \beta_R + u_i + e_i$$
$$y = X\beta_R + Zu + e$$

Correlated Effects

Study	y_1	y_2	v_{11}	v_{21}	v_{22}
1	0.25	0.45	0.02	0.01	0.01
2	0.34	0.35	0.03	NA	0.02
3	0.43	NA	0.02	NA	NA
4	NA	0.27	NA	NA	0.01

Note. X and Z are selection matrices (not all studies will provide all effect sizes).

Multivariate Meta-Analysis (MVMA)

Multivariate random-effects model (MVREM)

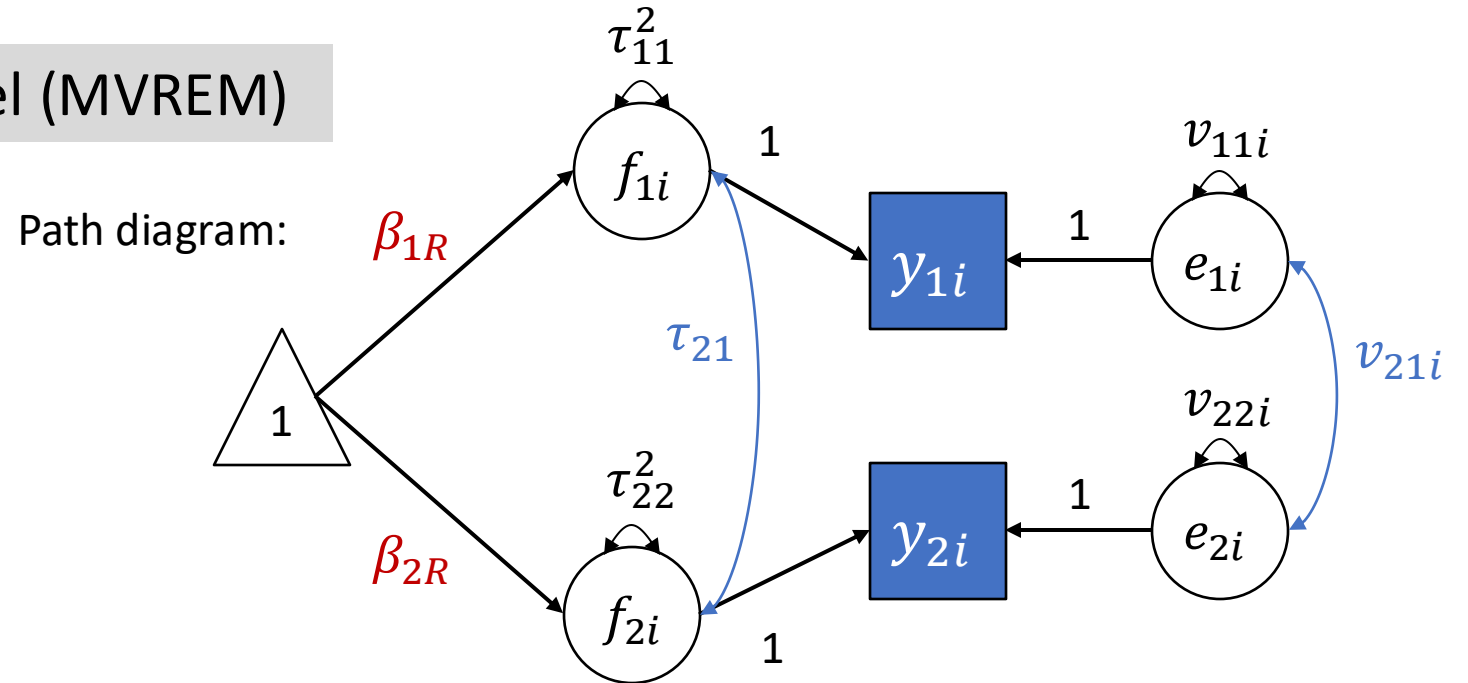
Part 1: $\mathbf{y}_i = \mathbf{f}_i + \mathbf{e}_i$
 $\mathbf{e}_i \sim MVN(\mathbf{0}, \mathbf{V}_i)$

Part 2: $\mathbf{f}_i = \boldsymbol{\beta}_R + \mathbf{u}_i$
 $\mathbf{u}_i \sim MVN(\mathbf{0}, \mathbf{T}^2)$

Total:

$$\mathbf{y}_i = \boldsymbol{\beta}_R + \mathbf{u}_i + \mathbf{e}_i$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_R + \mathbf{Z}\mathbf{u} + \mathbf{e}$$



Weighted average population
effect size under the MVREM

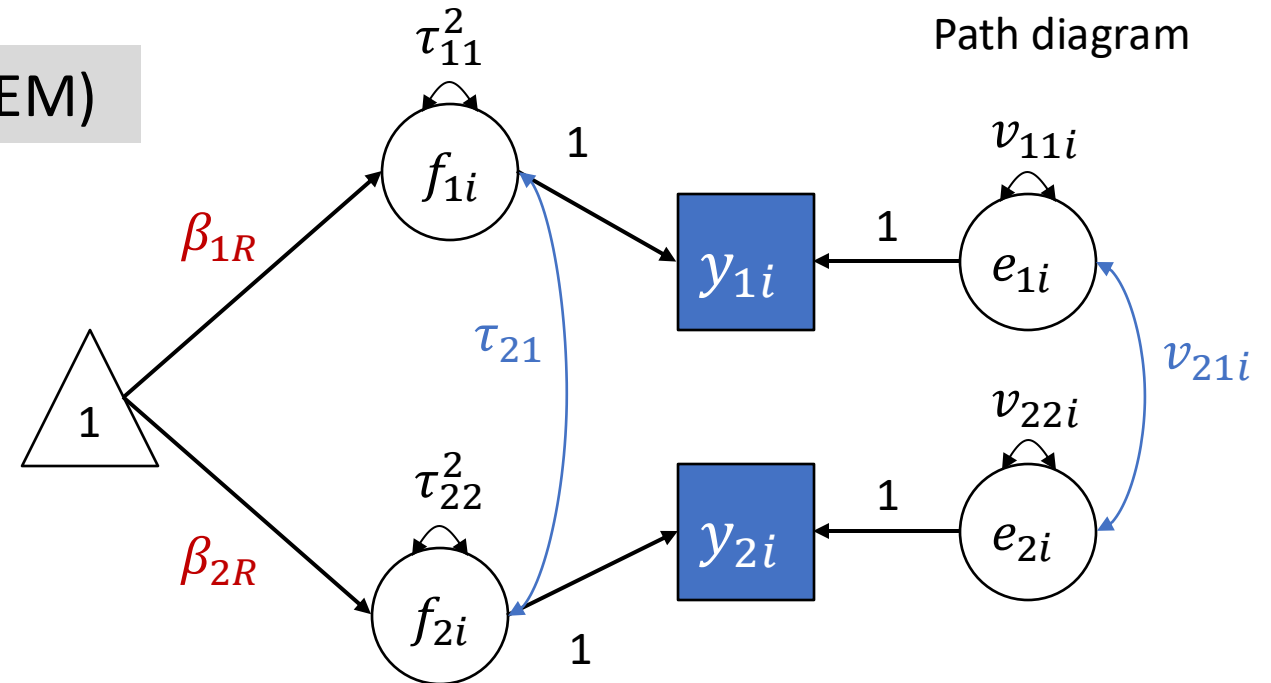
$$\boldsymbol{\beta}_R = \begin{bmatrix} \beta_{1R} \\ \beta_{2R} \end{bmatrix}$$

Multivariate Meta-Analysis (MVMA)

Multivariate **random-effects** model (MVREM)

Some notes on the MVREM:

- Different structures of the covariance matrix of random effects possible ($\mathbf{T}_{UN}^2$, $\mathbf{T}_{Diag}^2$, $\mathbf{T}_{CS}^2$, $\mathbf{T}_{Zero}^2$)
- Sampling covariances v_{21i} are often unknown → **SENSITIVITY ANALYSES**
- MVREM preferred over univariate REM
- Extensions to MASEM possible



$$\boldsymbol{\beta}_R = \begin{bmatrix} \beta_{1R} \\ \beta_{2R} \end{bmatrix}$$



Data Example: Mindfulness Interventions

Meta-analysis of the Effectiveness of Mindfulness Interventions

<https://doi.org/10.1037/bul0000359>

Tran et al. (2022)

- Effect sizes as Cohen's d
- Two outcomes: Self-reported mindfulness and mental health
146 RCTs with 292 effects
- Assumption: $r = 0.35$ between effects



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ISSN: 0033-2909

Psychological Bulletin

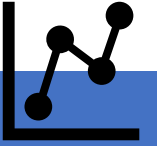
2022, Vol. 148, Nos. 1–2, 86–106
<https://doi.org/10.1037/bul0000359>

Self-Reported Mindfulness Accounts for the Effects of Mindfulness Interventions and Nonmindfulness Controls on Self-Reported Mental Health: A Preregistered Systematic Review and Three-Level Meta-Analysis of 146 Randomized Controlled Trials

Ulrich S. Tran, Layla Birnbaum, Matthias A. Burzler, Ulrich J. C. Hegewisch,
Dariga Ramazanova, and Martin Voracek

Department of Cognition, Emotion, and Methods in Psychology, School of Psychology, University of Vienna

```
## Calculate the sampling covariance matrix  
rho <- 0.35  
dat$SampCov <- sqrt(dat$vardmh)*sqrt(dat$vardmin)*rho
```



Data Example: Mindfulness Interventions

Multivariate Fixed-Effects Model

```
## Model estimation
MVFEM <- metaSEM::meta(y = cbind(dmh, dmin),
  v = cbind(vardmh, SampCov, vardmin),
  data = dat,
  model.name = "Multivariate FEM",
  RE.constraints = matrix(0, ncol=2, nrow=2))

MVFEM <- metaSEM::rerun(MVFEM)
```

No heterogeneity of effect
sizes assumed

```
## 95% confidence intervals: z statistic approximation (robust=FALSE)
## Coefficients:
##           Estimate Std. Error   lbound   ubound z value Pr(>|z|)
## Intercept1 0.463825   0.012493 0.439340 0.488310  37.128 < 2.2e-16 ***
## Intercept2 0.421650   0.014076 0.394061 0.449238  29.955 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Q statistic on the homogeneity of effect sizes: 1021.141
## Degrees of freedom of the Q statistic: 290
## P value of the Q statistic: 0
##
## Heterogeneity indices (based on the estimated Tau2):
##                               Estimate
## Intercept1: I2 (Q statistic)         0
## Intercept2: I2 (Q statistic)         0
```

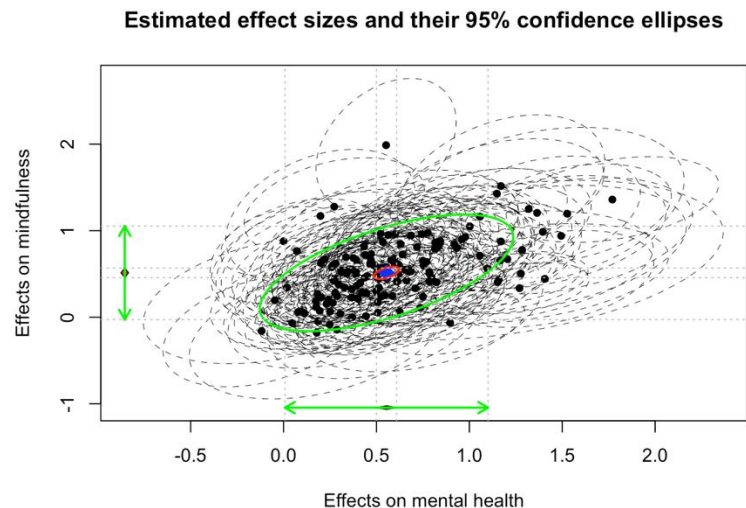


Data Example: Mindfulness Interventions

Multivariate Random-Effects Model

```
## Model estimation
MVREM <- metaSEM::meta(y = cbind(dmh, dmin),
                        v = cbind(vardmh, SampCov, vardmin),
                        data = dat,
                        model.name = "Multivariate REM")

MVREM <- metaSEM::rerun(MVREM)
```



```
## Coefficients:
##           Estimate Std.Error   lbound   ubound z value Pr(>|z|)
## Intercept1 0.554217  0.027767  0.499795  0.608639 19.9596 < 2.2e-16 ***
## Intercept2 0.512777  0.028414  0.457087  0.568466 18.0468 < 2.2e-16 ***
## Tau2_1_1    0.077740  0.013328  0.051618  0.103861  5.8330 5.445e-09 ***
## Tau2_2_1    0.047061  0.010507  0.026467  0.067654  4.4789 7.502e-06 ***
## Tau2_2_2    0.075858  0.013382  0.049629  0.102087  5.6685 1.440e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Q statistic on the homogeneity of effect sizes: 1021.141
## Degrees of freedom of the Q statistic: 290
## P value of the Q statistic: 0
##
## Heterogeneity indices (based on the estimated Tau2):
##                               Estimate
## Intercept1: I2 (Q statistic)  0.7718
## Intercept2: I2 (Q statistic)  0.7221
```

Heterogeneity assumed



Data Example: Mindfulness Interventions

Multivariate Mixed-Effects Meta-Regression Model

```
## Model estimation
MVMEM.linear <- metaSEM::meta(y = cbind(dmh, dmin),
                             v = cbind(vardmh, SampCov, vardmin),
                             data = dat,
                             x = dat$zRoBsum,
                             R2 = TRUE,
                             model.name = "Linear moderator effects")

MVMEM.linear <- metaSEM::rerun(MVMEM.linear)
```

Moderation by the risk of bias score for each study (RoB 2 assessment), assuming linear and freely estimated moderator effects

```
## Coefficients:
##           Estimate Std.Error    lbound    ubound z value Pr(>|z|)
## Intercept1  0.555967  0.027664  0.501746  0.610188 20.0969 < 2.2e-16 ***
## Intercept2  0.514535  0.028433  0.458808  0.570262 18.0966 < 2.2e-16 ***
## Slope1_1    0.029412  0.027647 -0.024775  0.083599  1.0638  0.2874
## Slope2_1    0.026396  0.028545 -0.029552  0.082344  0.9247  0.3551
## Tau2_1_1    0.076556  0.013231  0.050625  0.102488  5.7862 7.199e-09 ***
## Tau2_2_1    0.046130  0.010429  0.025690  0.066570  4.4233 9.722e-06 ***
## Tau2_2_2    0.075249  0.013306  0.049169  0.101329  5.6551 1.557e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Q statistic on the homogeneity of effect sizes: 1021.141
## Degrees of freedom of the Q statistic: 290
## P value of the Q statistic: 0
##
## Explained variances (R2):
##                               y1      y2
## Tau2 (no predictor)      0.077740 0.0759
## Tau2 (with predictors) 0.076556 0.0752
## R2                      0.015220 0.0080
```


Meta-Analytic Structural Equation Modeling (MASEM)

Meta-Analytic Structural Equation Modeling (MASEM)

When is MASEM useful?

- Test one or more theories on the relations among variables via SEM
- Reliability and validity generalization
- Test heterogeneity in specific SEM parameters (e.g., indirect effects)
- Evaluate models with latent variables (e.g., measurement invariance testing)
- Integrative analysis of open-access data sets
- ...

Meta-Analytic Structural Equation Modeling (MASEM)

Meta-analysis of correlation matrices (i.e., multiple correlations from each study) or SEM parameters



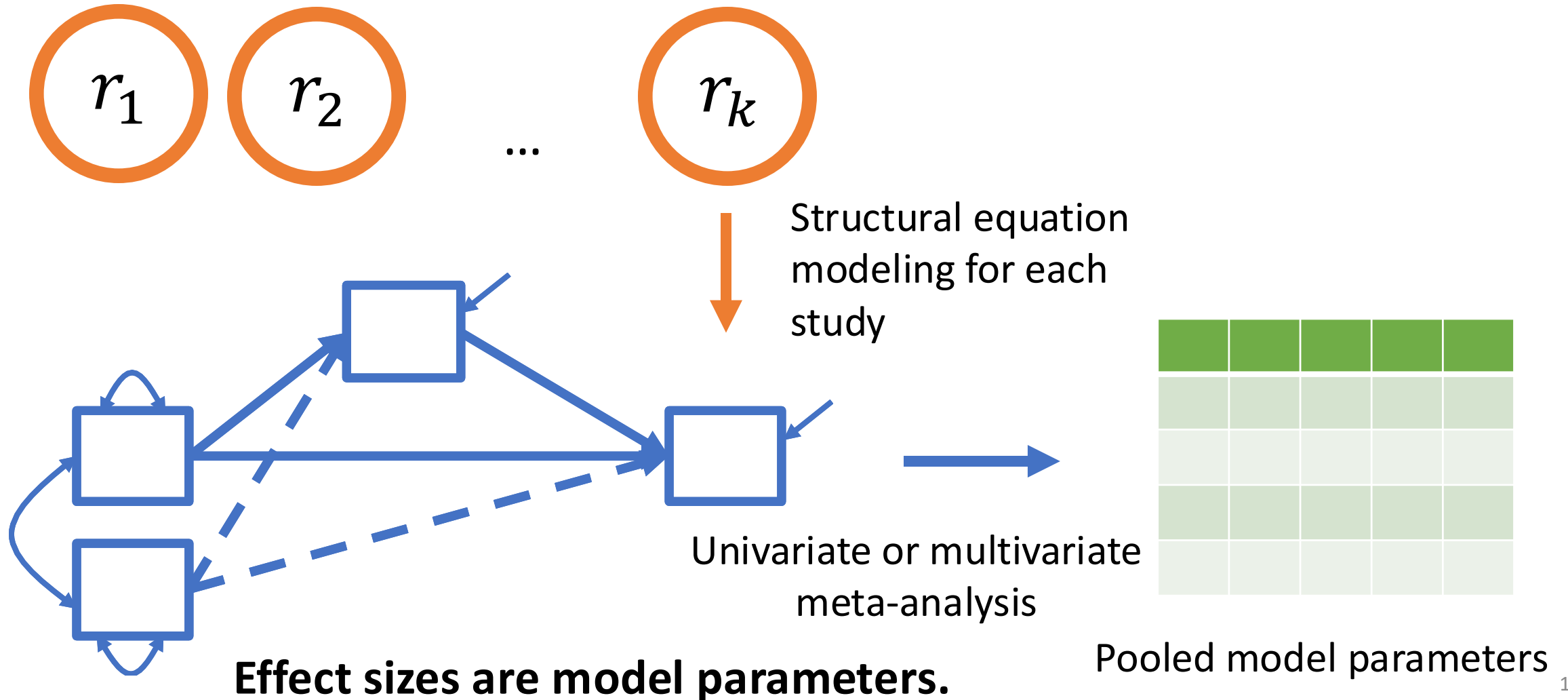
Parameter-based MASEM

- Stage 1: SEM for each study
- Stage 2: (Multivariate) meta-analysis of SEM parameters
- Quantify heterogeneity in SEM parameters

Correlation-based MASEM

- Synthesize correlation matrices to an overall correlation matrix with some heterogeneity
- One- or two-stage MASEM
- Test the fit of models

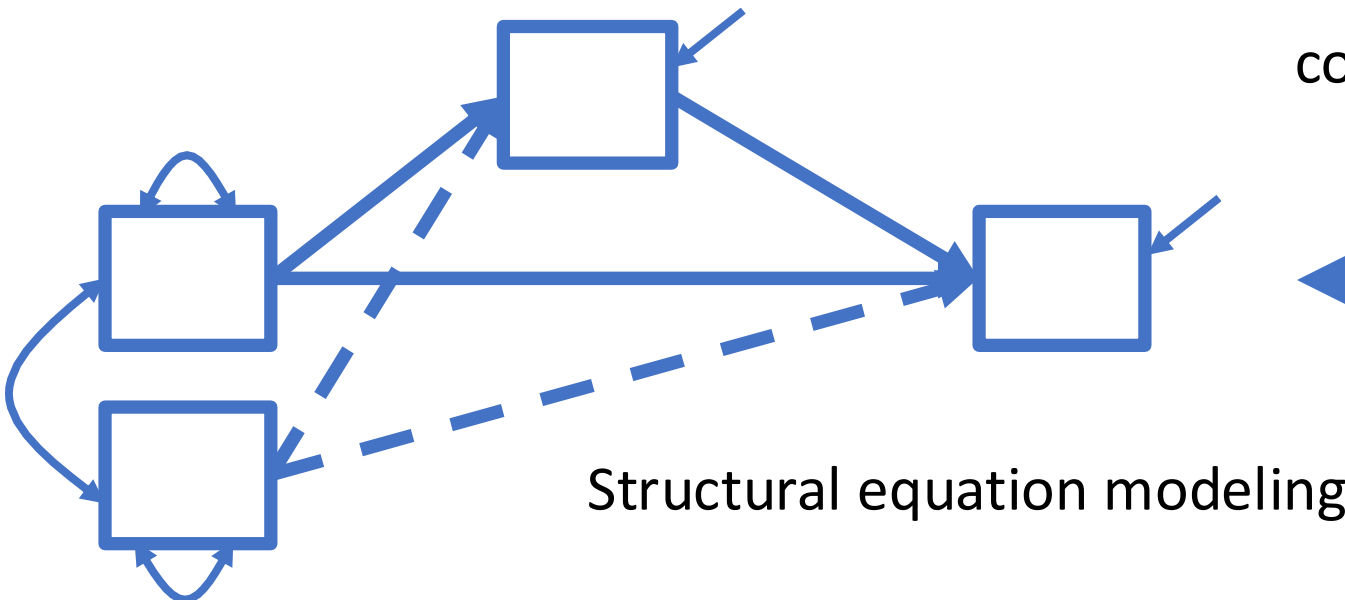
Parameter-Based MASEM



Correlation-Based MASEM

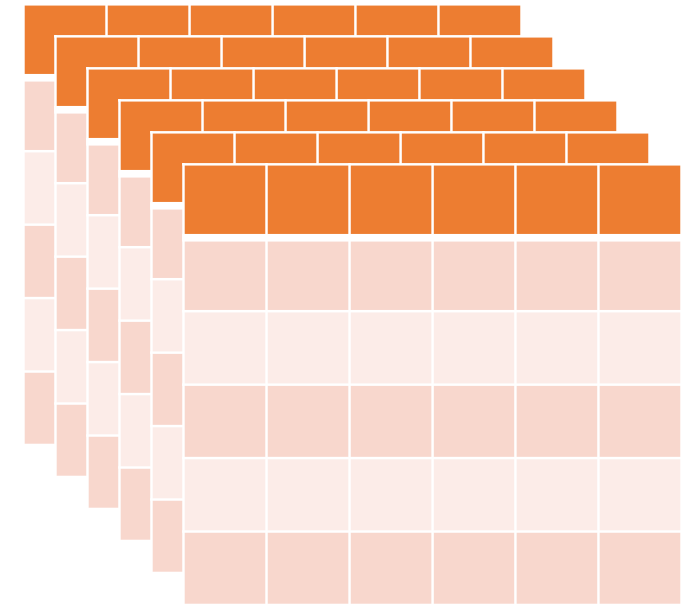


Effect sizes are correlation matrices.

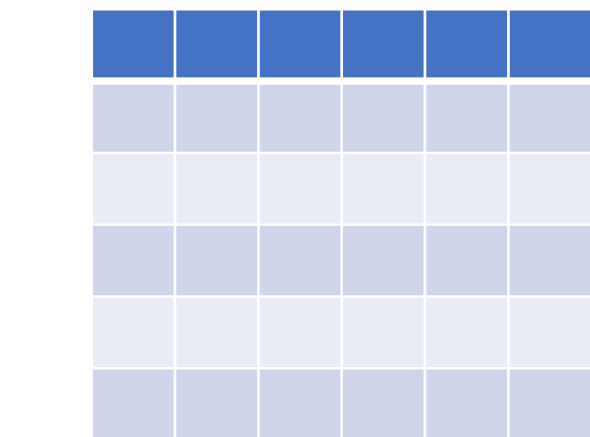


Structural equation modeling

Aggregating
correlation matrices



(Cheung, 2015)



Pooled correlation matrix

Correlation-Based MASEM

Synthesizing correlation matrices meta-analytically

Correlation-Based MASEM: Typical Data Set

Study	r_{21}	r_{31}	r_{41}	r_{32}	r_{42}	r_{43}
Study 1	x	x	x	x	x	x
Study 2	x	x		x		
Study 3				x	x	x
Study 4	x	x	x	x	x	
...						
Study k	x	x	x	x	x	x

Data: Correlation matrices from primary studies (with missings)

Correlation-Based MASEM

Study	r_{21}	r_{31}	r_{41}	r_{32}	r_{42}	r_{43}
Study 1	x	x	x	x	x	x
Study 2	x	x		x		
Study 3				x	x	x
Study 4	x	x	x	x	x	
...						
Study k	x	x	x	x	x	x

Stage **1**

Pooling correlation matrices:
Maximum likelihood estimation

Pooled correlations and variance components:

$$\mu(P) = (\rho_{21}, \rho_{31}, \rho_{41}, \rho_{32}, \rho_{42}, \rho_{43})$$

$$\Sigma(P) = V_i + T^2$$

$Acov$: Asymptotic covariance matrix of pooled correlations

Two-Stage Correlation-based MASEM (TSSEM)

Stage 1: Pooling correlation matrices (multi-group SEM)

Multivariate random-effects model

(Cheung, 2015; Cheung & Cheung, 2016)

Level 1:

$$\mathbf{r}_i = \boldsymbol{\rho}_i + \mathbf{e}_i$$

Level 2:

$$\boldsymbol{\rho}_i = \boldsymbol{\rho}_R + \mathbf{u}_i$$



Vector of the pooled correlation matrix $\hat{\boldsymbol{\rho}}_R$
Asymptotic sampling covariance matrix $\hat{\mathbf{V}}_R$
Heterogeneity of correlation vectors $\hat{\mathbf{T}}^2$

$$\mathbf{e}_i \sim N(\mathbf{0}, \mathbf{V}_i) \text{ and } \mathbf{u}_i \sim N(\mathbf{0}, \mathbf{T}^2)$$

Sampling and heterogeneity covariance matrices

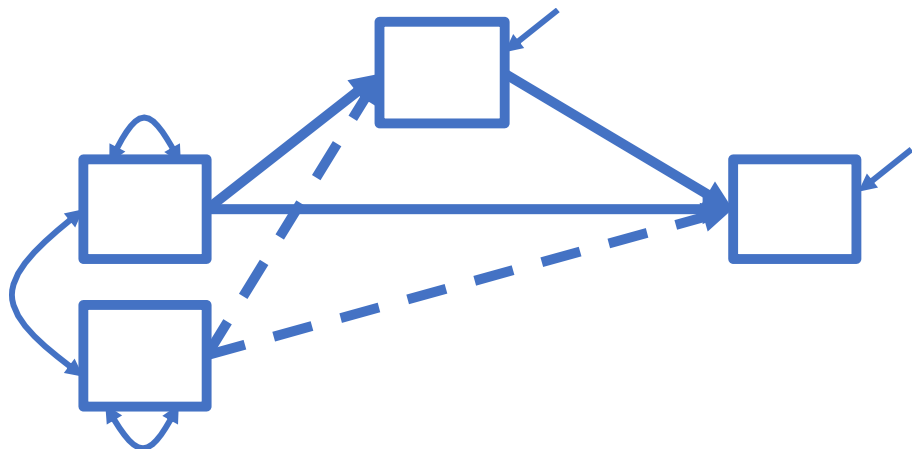
Correlation-Based MASEM

Pooled correlation matrix:

1.00			
ρ_{21}	1.00		
ρ_{31}	ρ_{32}	1.00	
ρ_{41}	ρ_{42}	ρ_{43}	1.00

Stage **2**

Pooled correlations as
observed input matrix



SEM through weighted least
squares (WLS) or maximum-
likelihood (ML) estimation

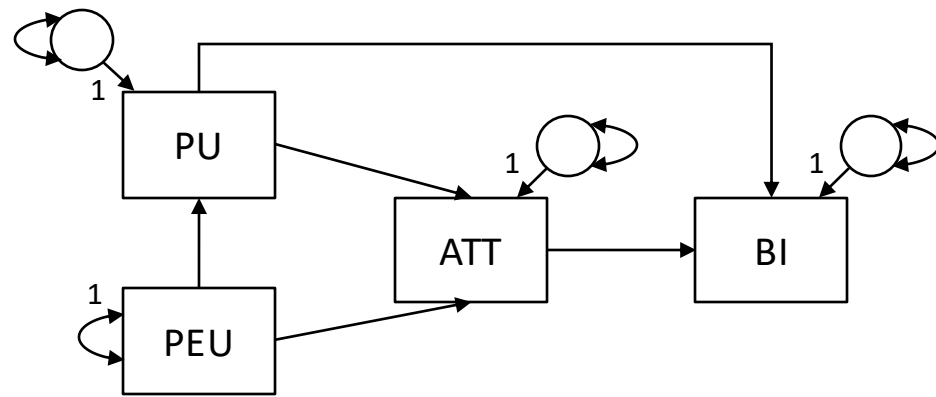


Technology Acceptance Model (TAM)

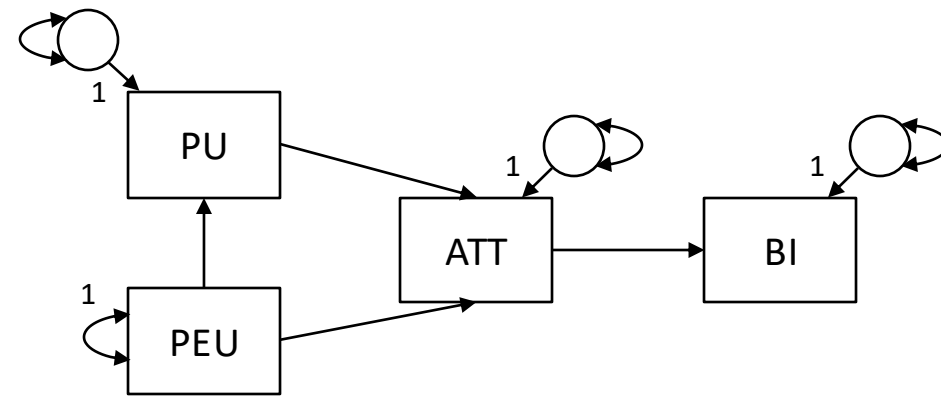
A Model Explaining Teachers' Intentions to Use Technology

Variables

- BI: Intentions to use technology, ATT: Attitudes toward technology
- PEU: Perceived ease of use, PU: Perceived usefulness



Model 1



Model 2



Technology Acceptance Model (TAM)

Meta-analytic data set based on teacher samples

Data set

- 50 studies with independent samples
- 4 variables
- 300 correlations
- 50 correlation matrices
- Pre- and in-service teacher samples

Main RQs

- 1. Model evaluation:** To what extent does the TAM represent the teacher data across samples?
- 2. Model comparison:** To what extent does a direct effect exist?

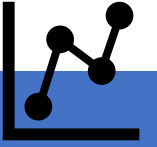
⇒ Correlation-based MASEM



Technology Acceptance Model (TAM)

Meta-analytic data set based on teacher samples

	A	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH
1	StudyID	Sample size	PU-PU	PU-PEOU	PEOU	PU-ATT	PEOU-ATT	ATT-ATT	PU-BI	PEOU-BI	ATT-BI	BI-BI
2	217	446	1	0,61	1	0,6776	0,5138	1	0,599184	0,449892	0,860328	1
3	247	88	1	0,661	1	0,59	0,584	1	0,75	0,746	0,684	1
4	281	271	1	0,89	1	0,423	0,187	1	0,374	0,82	0,512	1
5	289	234	1	0,5	1	0,64	0,55	1	0,55	0,48	0,59	1
6	311	29	1	0,512	1	0,461	0,754	1	0,364	0,669	0,595	1
7	320	313	1	0,279	1	0,63	0,873	1	0,319	0,441	0,505	1
8	332A	169	1	0,68	1	0,68	0,73	1	0,34	0,3	0,3	1
9	332B	170	1	0,4	1	0,56	0,43	1	0,23	0,32	0,36	1
10	335A	151	1	0,329	1	0,565	0,349	1	0,523	0,327	0,462	1



Technology Acceptance Model (TAM)

Step 0

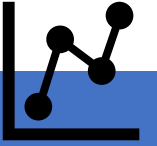
Check correlation matrices
for their positive
definiteness

```
# PD check  
pd_check <- is.pd(TAMu$data)  
sum(!is.na(pd_check))
```

```
## [1] 50
```

```
pd_check
```

```
## Okazaki & Renda dos Santos (2012)  
## TRUE  
## Wong (2015)  
## TRUE
```



Technology Acceptance Model (TAM)

Two-stage MASEM

Step 1

Pooling correlation matrices across independent samples

- REM
- Diagonal constraints of the heterogeneity matrix

```
# Combine correlation matrices using a REM  
#  
rem <- tssem1(TAMu$data, TAMu$n,  
              method = "REM",  
              RE.type = "Diag")  
  
# Summarize the results  
summary(rem)
```



Technology Acceptance Model (TAM)

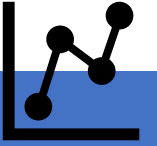
Two-stage MASEM

Step 1

Pooling correlation matrices across independent samples

- REM
- Diagonal constraints of the heterogeneity matrix

```
## 95% confidence intervals: z statistic approximation (robust=FALSE)
## Coefficients:
##              Estimate Std.Error    lbound    ubound z value Pr(>|z|)
## Intercept1  0.4828802  0.0217706  0.4402105  0.5255498 22.1803 < 2.2e-16 ***
## Intercept2  0.6068160  0.0174683  0.5725787  0.6410533 34.7381 < 2.2e-16 ***
## Intercept3  0.5153033  0.0254332  0.4654551  0.5651514 20.2611 < 2.2e-16 ***
## Intercept4  0.5276429  0.0209277  0.4866253  0.5686604 25.2127 < 2.2e-16 ***
## Intercept5  0.4089908  0.0220210  0.3658303  0.4521512 18.5727 < 2.2e-16 ***
## Intercept6  0.5187087  0.0259696  0.4678093  0.5696081 19.9737 < 2.2e-16 ***
## Tau2_1_1    0.0210148  0.0047131  0.0117772  0.0302524  4.4588 8.243e-06 ***
## Tau2_2_2    0.0134545  0.0030663  0.0074446  0.0194643  4.3878 1.145e-05 ***
## Tau2_3_3    0.0298527  0.0064754  0.0171612  0.0425443  4.6102 4.023e-06 ***
## Tau2_4_4    0.0195271  0.0044114  0.0108809  0.0281733  4.4265 9.578e-06 ***
## Tau2_5_5    0.0210140  0.0048452  0.0115176  0.0305103  4.3371 1.444e-05 ***
## Tau2_6_6    0.0312785  0.0066695  0.0182064  0.0443505  4.6898 2.735e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
....
```



Technology Acceptance Model (TAM)

Two-stage MASEM

Step 1

Pooling correlation matrices across independent samples

- REM
- Diagonal constraints of the heterogeneity matrix

```
## Q statistic on the homogeneity of effect sizes: 3622.948
## Degrees of freedom of the Q statistic: 294
## P value of the Q statistic: 0
##
## Heterogeneity indices (based on the estimated Tau2):
##                                     Estimate
## Intercept1: I2 (Q statistic)      0.9178
## Intercept2: I2 (Q statistic)      0.9146
## Intercept3: I2 (Q statistic)      0.9460
## Intercept4: I2 (Q statistic)      0.9217
## Intercept5: I2 (Q statistic)      0.9015
## Intercept6: I2 (Q statistic)      0.9493
```



Technology Acceptance Model (TAM)

Two-stage MASEM

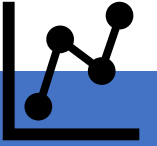
Step 1

Pooling correlation matrices across independent samples

- REM
- Diagonal constraints of the heterogeneity matrix

```
# Extract the pooled correlation matrix  
corr.rem <- vec2symMat(coef(rem, select="fixed"), diag=FALSE)  
colnames(corr.rem) <- c("PU", "PEU", "ATT", "BI")  
rownames(corr.rem) <- c("PU", "PEU", "ATT", "BI")  
corr.rem
```

##		PU	PEU	ATT	BI
##	PU	1.0000000	0.4828802	0.6068160	0.5153033
##	PEU	0.4828802	1.0000000	0.5276429	0.4089908
##	ATT	0.6068160	0.5276429	1.0000000	0.5187087
##	BI	0.5153033	0.4089908	0.5187087	1.0000000



Technology Acceptance Model (TAM)

Two-stage MASEM

Step 2

Model specification

- Lavaan syntax possible
- Translate into the RAM framework

```
# Model specification
```

```
Model1 <- " # Structural parameters
```

```
BI ~ ATT + PU
```

```
ATT ~ PEU + PU
```

```
PU ~ PEU
```

```
# Variance of independent variables
```

```
PEU ~~ 1*PEU
```

```
"
```



Technology Acceptance Model (TAM)

Two-stage MASEM

Step 2

Model specification

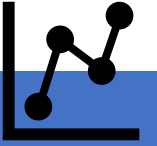
- Lavaan syntax possible
- Translate into the RAM framework

```
# Convert the model into the RAM language
RAM <- lavaan2RAM(Model1,
                    obs.variables = c("PU", "PEU", "ATT", "BI"))

# Declare the matrices in the RAM framework
# A: Matrix of factor loadings and regression coefficients
A <- RAM$A
# S: Matrix of variances and covariances
S <- RAM$S
# F: Matrix indicating manifest or latent variables (0=latent, 1=manifest)
F <- RAM$F

# Check the RAM specification
# Note: The function remains silent if the specification is correct.
checkRAM(A, S, cor.analysis = TRUE)
```

MetaSEM uses the **R**ecticular **A**ction **M**odel notation to specify structural equation models (McArdle, 2005).



Technology Acceptance Model (TAM)

Two-stage MASEM

Step 2

Model estimation

- Pooled correlation matrix and model-specifying matrices as input
- Diagonal constraints

```
# Model estimation  
# Confidence intervals are likelihood-ratio based (LBCIs)  
TS.Model1 <- tssem2(rem, Amatrix=A, Smatrix=S, Fmatrix=F,  
                    intervals.type="LB",  
                    diag.constraints = TRUE,  
                    model.name = "Model 1 with the direct effect")
```




Technology Acceptance Model (TAM)

Two-stage MASEM

Step 2

Model estimation

- Pooled correlation matrix and model-specifying matrices as input
- Diagonal constraints

```
## 95% confidence intervals: Likelihood-based statistic
## Coefficients:
##              Estimate Std.Error   lbound   ubound z value Pr(>|z|)
## ATTONPEU      0.32668         NA 0.26635 0.38442      NA      NA
## ATTONPU       0.43668         NA 0.38557 0.48754      NA      NA
## BIONATT       0.35455         NA 0.26355 0.44470      NA      NA
## BIONPU        0.32253         NA 0.23013 0.41347      NA      NA
## PUONPEU       0.49973         NA 0.45797 0.54132      NA      NA
## ATTWITHATT    0.56001         NA 0.51662 0.60096      NA      NA
## BIWITHBI      0.63306         NA 0.58739 0.67590      NA      NA
## PUWITHPU      0.75027         NA 0.70697 0.79026      NA      NA
```

```
# Summarize the model parameters
summary(TS.Model1)
```



Technology Acceptance Model (TAM)

Two-stage MASEM

Step 2

Model estimation

- Pooled correlation matrix and model-specifying matrices as input
- Diagonal constraints

```
# Summarize the model parameters  
summary(TS.Model1)
```

```
## Goodness-of-fit indices:  
##                                     Value  
## Sample size                        14918.0000  
## Chi-square of target model          10.0900  
## DF of target model                  1.0000  
## p value of target model             0.0015  
## Number of constraints imposed on "Smatrix" 3.0000  
## DF manually adjusted                0.0000  
## Chi-square of independence model     3002.8440  
## DF of independence model            6.0000  
## RMSEA                              0.0247  
## RMSEA lower 95% CI                  0.0125  
## RMSEA upper 95% CI                  0.0395  
## SRMR                               0.0285  
## TLI                                0.9818  
## CFI                                0.9970  
## AIC                                8.0900  
## BIC                                0.4797
```



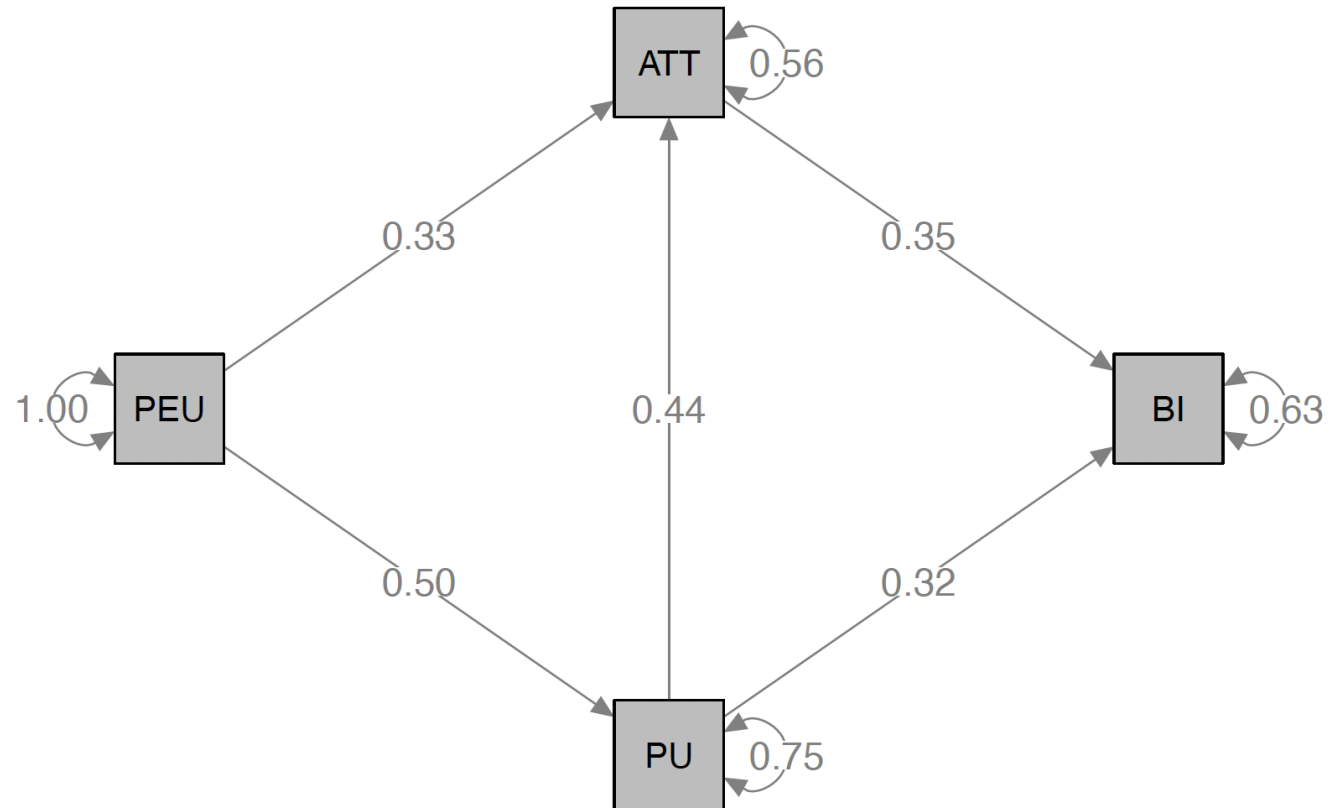
Technology Acceptance Model (TAM)

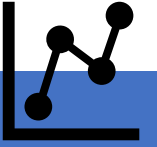
Two-stage MASEM

Step 2

Path diagram

```
# Visualize the model and add its parameters
TS.Model1.plot <- meta2semPlot(TS.Model1)
semPaths(TS.Model1.plot, whatLabels="est",
  rotation=2,
  edge.label.cex = 1.25,
  sizeMan = 8,
  color = "grey",
  layout = "tree2")
```





Technology Acceptance Model (TAM)

Two-stage MASEM

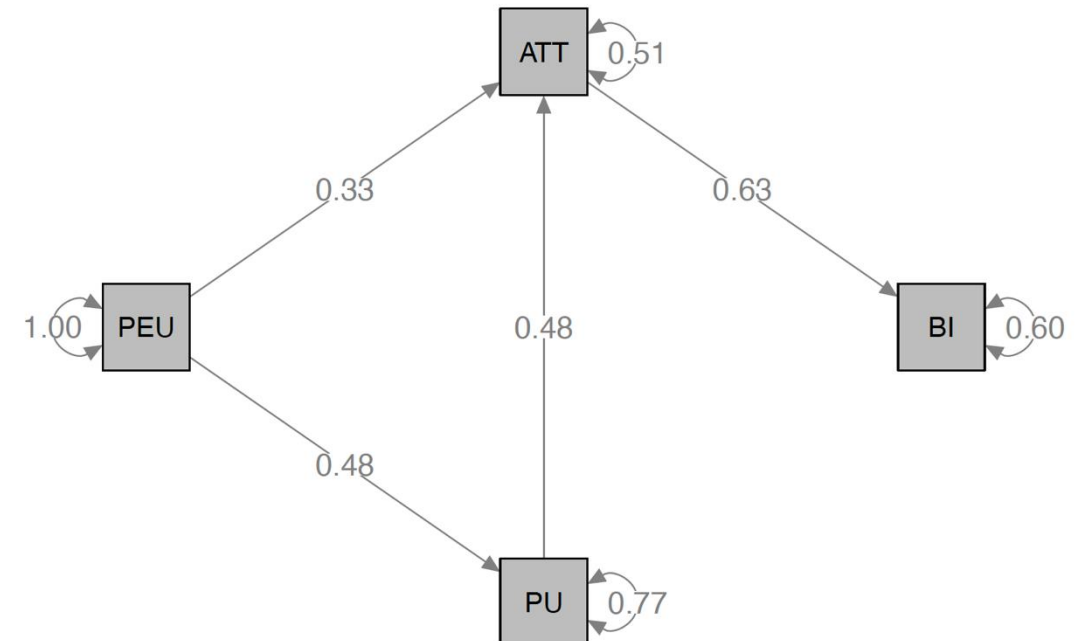
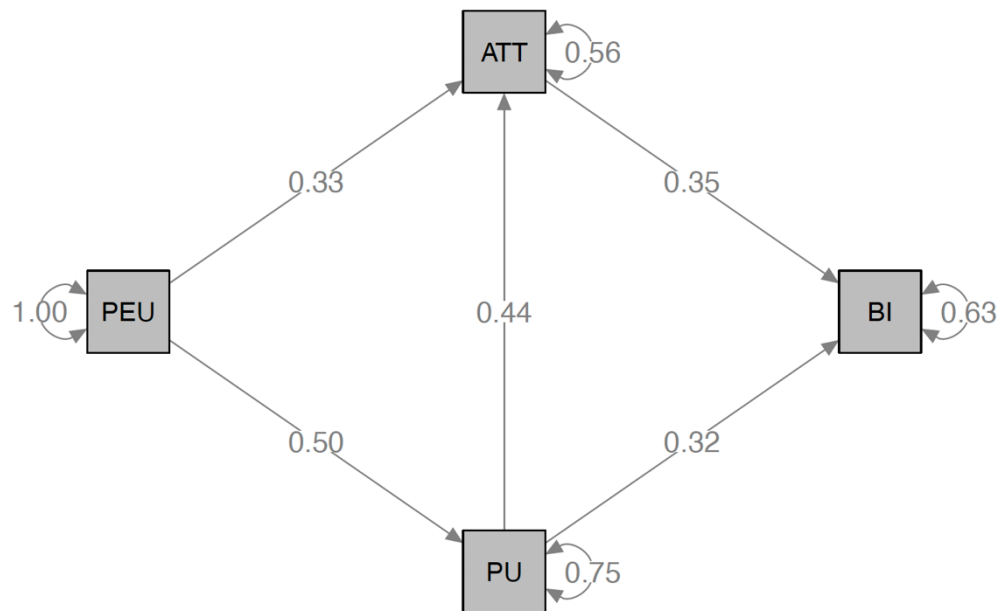
Step 3

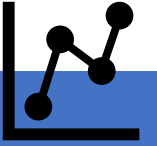
Model comparison

```
# Model comparison
```

```
anova(TS.Model1, TS.Model2)
```

Likelihood-ratio test: $\chi^2(1) = 42.1$, $p < .001$





Technology Acceptance Model (TAM)

One-stage MASEM

Steps 1 & 2

Pooling correlation matrices and estimating the SEM at the same time

```
tamos <- Cor2DataFrame(TAMu$data, TAMu$n)
```

```
## Pooling of correlation matrices and model estimation  
M1_os <- osmasem(RAM = RAM,  
                 model.name = "Model 1 OSMASEM",  
                 data = tamos)
```

```
## Summarize the results  
summary(M1_os)
```

```
## Pooling of correlation matrices and model estimation  
M2_os <- osmasem(RAM = RAM2,  
                 model.name = "Model 2 OSMASEM",  
                 data = tamos)
```

```
## Summarize the results  
summary(M2_os)
```

Add `fitIndices = TRUE` to the `summary()` function to get the goodness-of-fit indices.



Technology Acceptance Model (TAM)

One-stage MASEM

Steps 1 & 2
Pooling correlation
matrices and
estimating the SEM at
the same time

```
## Summary of Model 1 OSMASEM
```

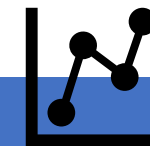
```
##
```

```
## free parameters:
```

##		name	matrix	row	col	Estimate	Std.Error	A	z value	Pr(> z)
## 1	ATTONPU	A0	ATT	PU	0.4387429	0.02644554			16.590432	0.000000e+00
## 2	BIONPU	A0	BI	PU	0.3217157	0.04784739			6.723788	1.770606e-11
## 3	PUONPEU	A0	PU	PEU	0.4982821	0.02179327			22.864033	0.000000e+00
## 4	ATTONPEU	A0	ATT	PEU	0.3247803	0.03083009			10.534525	0.000000e+00
## 5	BIONATT	A0	BI	ATT	0.3527154	0.04782889			7.374527	1.649791e-13
## 6	Tau1_1	vecTau1		1	1	-1.9209330	0.11309633		-16.984929	0.000000e+00
## 7	Tau1_2	vecTau1		2	1	-2.1517208	0.11428478		-18.827710	0.000000e+00
## 8	Tau1_3	vecTau1		3	1	-1.7457434	0.10938516		-15.959599	0.000000e+00
## 9	Tau1_4	vecTau1		4	1	-1.9578490	0.11418644		-17.146072	0.000000e+00
## 10	Tau1_5	vecTau1		5	1	-1.8521917	0.11779238		-15.724206	0.000000e+00
## 11	Tau1_6	vecTau1		6	1	-1.7164560	0.10860149		-15.805087	0.000000e+00

```
## Pooling of correlation matrices and model estimation
M1_os <- osmasem(RAM = RAM,
                 model.name = "Model 1 OSMASEM",
                 data = tamos)
```

```
## Summarize the results
summary(M1_os)
```



Technology Acceptance Model (TAM)

One-stage MASEM

Steps 1 & 2

Pooling correlation matrices and estimating the SEM at the same time

```
## Pooling of correlation matrices and model estimation
M1_os <- osmasem(RAM = RAM,
  model.name = "Model 1 OSMASEM",
  data = tamos)
```

```
## Summarize the results
summary(M1_os)
```

```
## SRMR
osmasemSRMR(M1_os)
```

```
## [1] 0.02843007
```

```
## Extract the model coefficients
coef(M1_os)
```

```
## ATTONPU BIONPU PUONPEU ATTONPEU BIONATT
## 0.4387429 0.3217157 0.4982821 0.3247803 0.3527154
```

```
## Heterogeneity covariance matrix
VarCorr(M1_os)
```

```
##           Tau2_1      Tau2_2      Tau2_3      Tau2_4      Tau2_5      Tau2_6
## Tau2_1 0.02145353 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
## Tau2_2 0.00000000 0.01352194 0.00000000 0.00000000 0.00000000 0.00000000
## Tau2_3 0.00000000 0.00000000 0.03045556 0.00000000 0.00000000 0.00000000
## Tau2_4 0.00000000 0.00000000 0.00000000 0.01992664 0.00000000 0.00000000
## Tau2_5 0.00000000 0.00000000 0.00000000 0.00000000 0.02461539 0.00000000
## Tau2_6 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.03229277
```




Technology Acceptance Model (TAM)

One-stage MASEM

Steps 1 & 2

Pooling correlation matrices and estimating the SEM at the same time

```
## Pooling of correlation matrices and model estimation
M1_os <- osmasem(RAM = RAM,
                 model.name = "Model 1 OSMASEM",
                 data = tamos)

## Summarize the results
summary(M1_os)
```

```
## Pooling of correlation matrices and model estimation
M2_os <- osmasem(RAM = RAM2,
                 model.name = "Model 2 OSMASEM",
                 data = tamos)

## Summarize the results
summary(M2_os)
```

```
## Model comparison
anova(M1_os, M2_os)
```

```
##           base      comparison ep  minus2LL  df      AIC    diffLL diffdf
## 1 Model 1 OSMASEM      <NA> 11 -231.4163 289 -209.4163      NA      NA
## 2 Model 1 OSMASEM Model 2 OSMASEM 10 -196.7256 290 -176.7256 34.69073      1
##           p
## 1          NA
## 2 3.864666e-09
```


The End.



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